

Scaling Hyperscale Cloud Networking from 400G to 800G

System Integration Lessons and Implications for OCP Platforms

OCP PLATINUM® Member

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1. Beyond SmartNICs: Networking solutions are servers in their own

Each generation now carries a full compute, storage, and security stack with software and firmware complexity comparable to the host server.

Networking and storage convergence

- A single device terminates the network data path and presents virtualized storage, offloading both from host CPUs at line rate.

Virtualized and bare-metal hosting

- The same platform serves VM fleets and bare-metal tenants, enforcing isolation and tenant policy independently of the host operating system.

ARM, x86, and SKU diversity

- AI, standard, and value SKUs span ARM and x86 cores, so one platform must absorb multiple silicon and feature combinations.

Security and full stack ownership

- Root of trust, attestation, secure provisioning, and a layered firmware and software stack make the NIC a managed server, not an adapter.

2. Industry challenge: bandwidth is no longer the only bottleneck

AI and hyperscale growth shift pressure to power, cooling, firmware, security, and validation.

What changes materially

- 400G front-end networking enables offload, low latency, high throughput, and energy efficiency.
- The 400G to 800G transition increases power density, thermal constraints, and integration coupling.
- Platform viability becomes a co-design problem across silicon, board, firmware, security, cooling, and validation.



Bandwidth scale | **Thermal** | **Power density** | **Validation scale**

3. Existing approaches and gaps

Reference-platform validation is necessary, but not sufficient for hyperscale deployment readiness.

Schedule alignment

- Silicon releases, firmware maturity, toolchain readiness, and platform milestones move at different velocities.

System observability

- Telemetry, debug tooling, and manufacturing diagnostics often arrive too late for efficient root cause analysis.

Configuration coverage

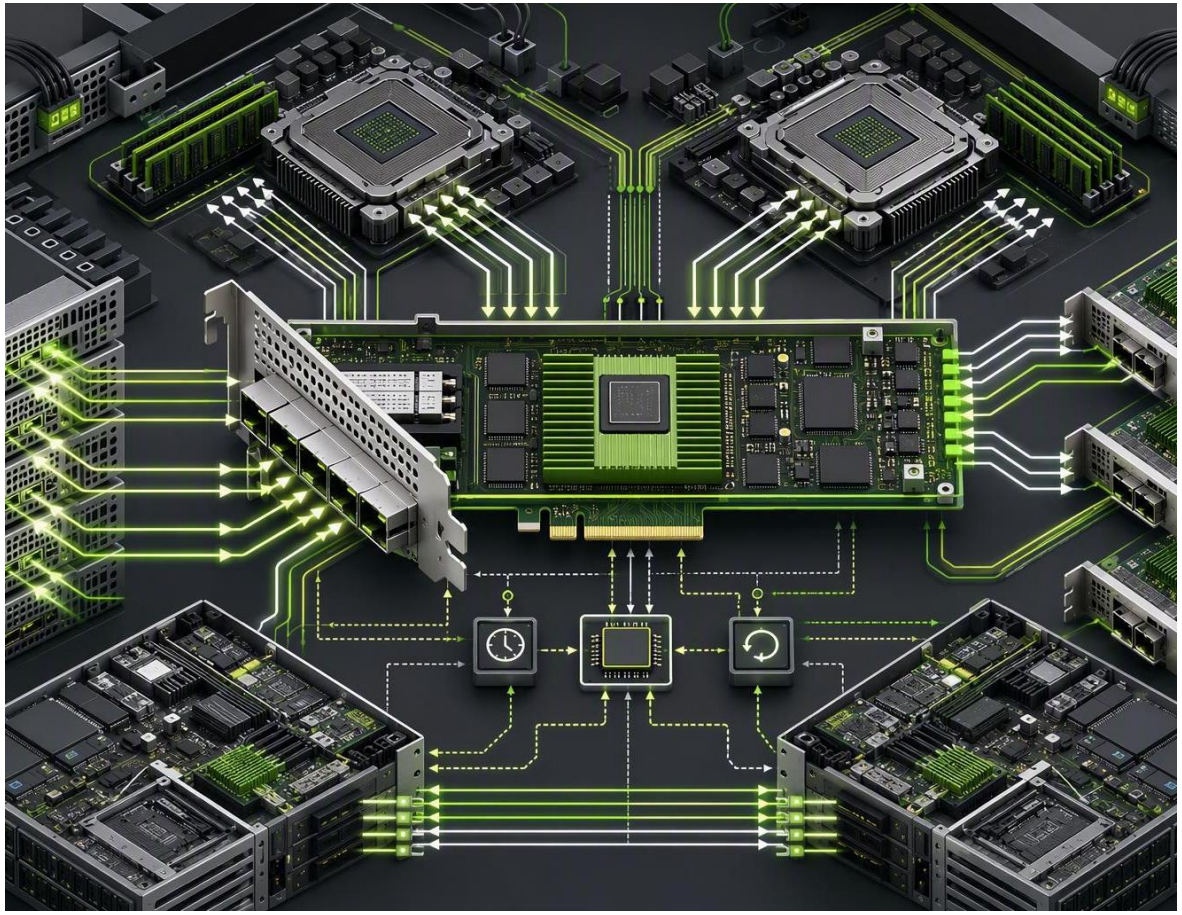
- Multi-host, multi-NIC, DDR5, liquid cooling, and security flows multiply validation permutations.

Ownership clarity

- Partner-side artifacts, platform deltas, and co-validation deliverables require explicit milestone tracking.

4. Proposed solution: deployment-grounded SmartNIC integration architecture

A reusable integration framework connecting architecture, firmware, telemetry, and validation milestones.

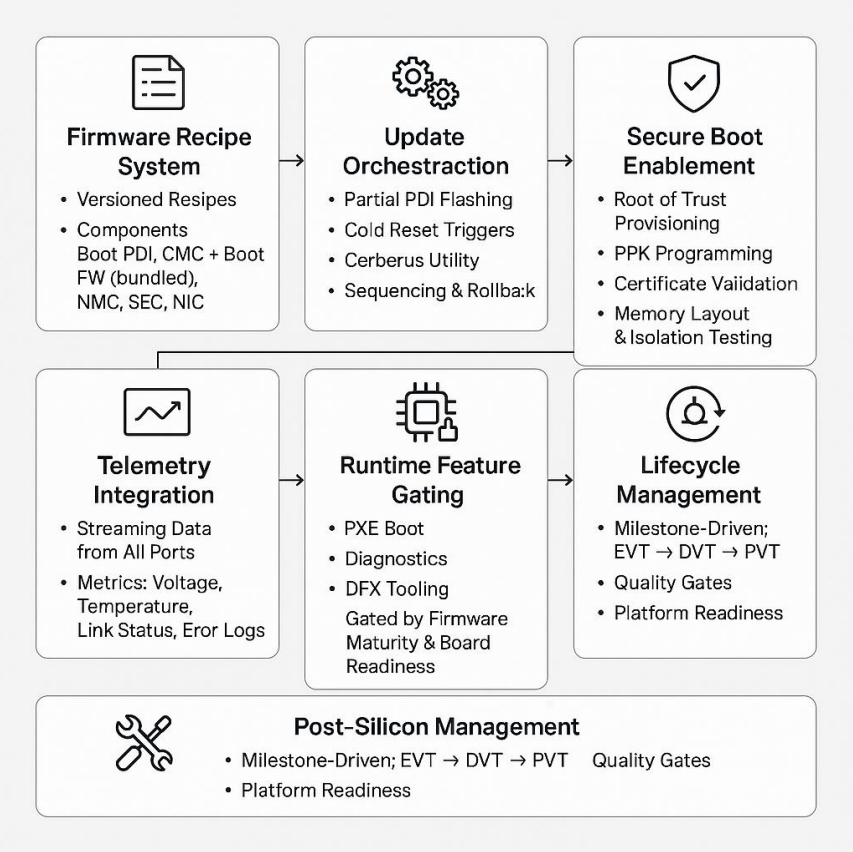


- Treat SmartNIC as a system-level acceleration layer, not an isolated adapter.
- Design for multi-generation compatibility with customized form factors where needed.
- Use shared milestones to align hyperscaler, silicon vendor, ODM, firmware, and manufacturing teams.
- Build observability and security provisioning into the architecture early.

Co-design **Security** **Telemetry**

5. Firmware, security, and telemetry as first-class integration deliverables

Scalable firmware recipes reduce regression risk and make security and observability repeatable.



*PDI (Programmable Device Image), CMC (Cerberus Management Controller), NMC (Network Management Controller), NIC (Network Interface Card) images, PPK (Platform Provisioning Key)

Firmware Integration Strategy

Modular Firmware Deployment

Security Boot & Telemetry

Milestone-Driven Update Process

Update Orchestration

Lifecycle & Post-Silicon Management

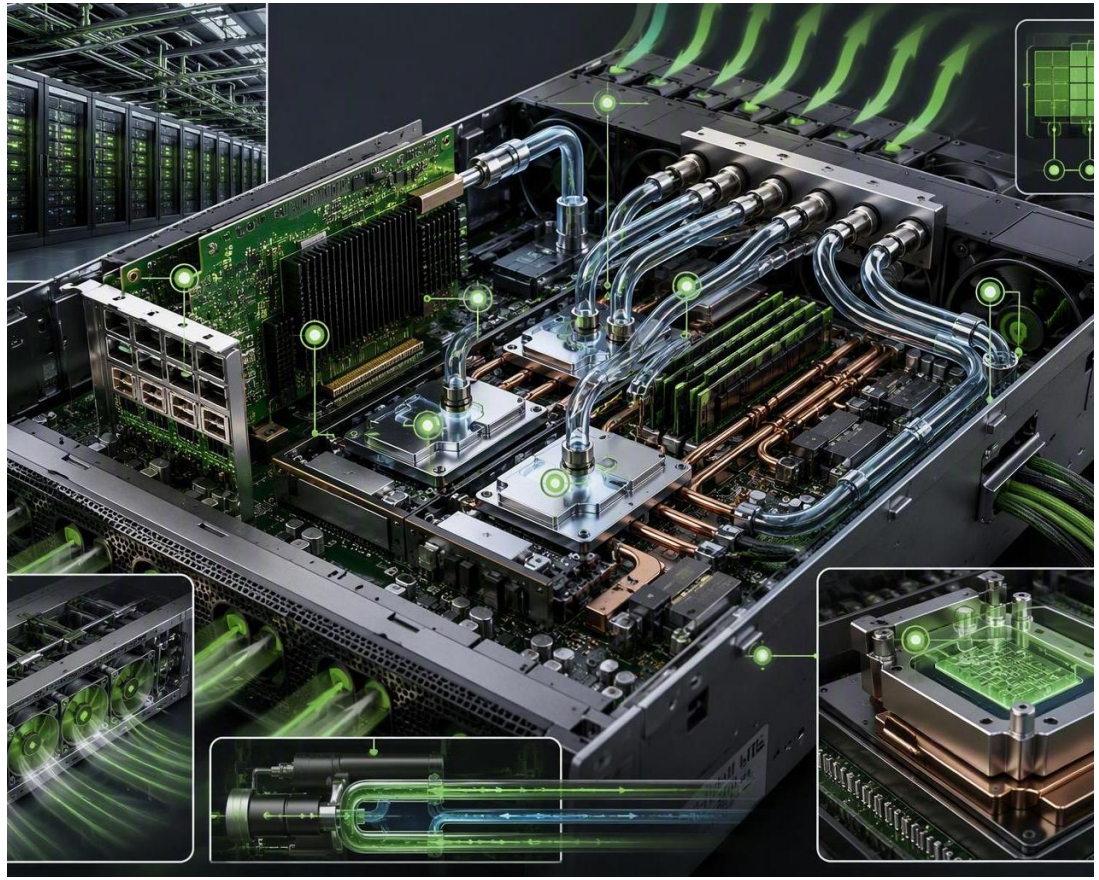
What changed since 100G: one vendor image became a multi-component versioned recipe, with provisioned security and streamed telemetry.

6. Key design details: multi-host, multi-NIC, memory, and thermal coupling

Performance scaling introduces electrical, mechanical, firmware, and cooling trade-offs.

800G platform scaling turns NIC integration into a system-level co-design problem

- **Multi-Host Architectures**: Scale host connectivity while preserving isolation and serviceability.
- **Multi-NIC Architectures**: Balance bandwidth growth, resiliency, and operational complexity.
- **PCIe Gen5/Gen6 Readiness**: Validate platform-level interoperability, recovery, and manageability.
- **System-Level Co-Design**: Align networking, memory, power, and mechanical constraints.
- **Thermal Sustainability**: Enable future power growth through liquid-ready designs.



7. Deployment and implementation experience

Successful 800G deployment requires validation across the full lifecycle, from ODM bring-up to production cluster integration.

01 **Silicon Readiness**

02 **ODM Bring-up & Validation**
(EVT/DVT/PVT)

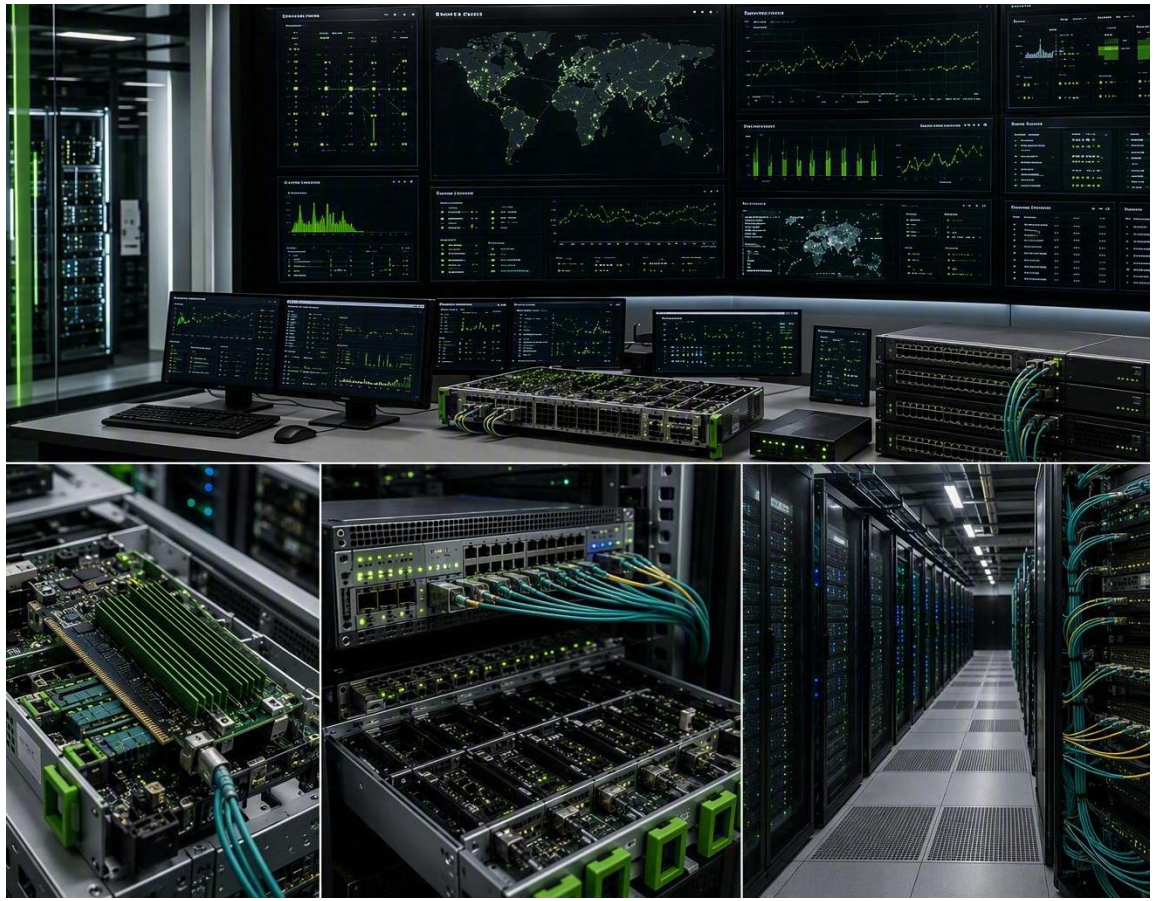
03 **Platform Integration & Rack Integration**
(BIOS/BMC/RSCM/FW/OS/Security)

04 **Build-Out & Rack Deployment**

05 **LSG Qualification**

06 **Cluster Docking & Service Validation**

07 **PV Exit & Production Readiness**



8. Results, benefits, and lessons learned

Results from slides 2-8: server-class NICs require lifecycle-ready integration, not adapter-only validation.

Benefits realized

- **Server-class NIC model** reframes 400G/800G devices as managed compute, storage, and security systems.
- **Reusable integration architecture** connects form factor, firmware, telemetry, security, and validation milestones.
- **Early system observability** improves root cause analysis across silicon, ODM, platform, and service teams.
- **Lifecycle readiness** creates a clearer path from bring-up to rack deployment, LSG, cluster docking, and PV exit.

Lessons learned

- **Bandwidth is not enough**: power, cooling, firmware, security, and validation now define platform viability.
- **Adapter-only validation is insufficient**; multi-host and multi-NIC designs must be validated at platform level.
- **Firmware, security, and telemetry** must be first-class deliverables with milestone-driven recipes and gates.
- **Co-design is mandatory** across silicon, board, PCIe, memory, thermal, mechanical, and operations.
- **Ownership clarity** is required for artifacts, deltas, recovery flows, and cross-stack deployment blockers.

OCP implications

- **Open lifecycle validation** should cover silicon readiness, ODM bring-up, platform/rack integration, and service validation.
- **Standard management artifacts** should include firmware recipes, secure provisioning, telemetry, diagnostics, and recovery.
- **Platform-level PCIe coverage** should address multi-host, multi-NIC, Gen5/Gen6 interoperability, recovery, and manageability.
- **Thermal-aware specifications** should support higher power density, memory coupling, and liquid-ready deployment.
- **Co-engineered ecosystems** should convert 400G lessons into reusable 800G-ready OCP platform practices.

9. Conclusion and discussion

From 400G deployment lessons to 800G-ready open platform practices.

Key takeaway

Future platform success will be determined not only by faster networking silicon, but by the ability to co-engineer compute, networking, cooling, management, and operations as a unified system.

Success increasingly depends on balancing:

- Scale drives complexity
- Power and cooling shape architecture
- Simplicity enables operations
- Security must be built in
- Deployment starts at design
- Ecosystem alignment accelerates adoption

Call to action

Develop a community contribution that captures reusable **design and validation practices for 800G-ready server platforms**.

Candidate deliverables

- System design guidelines
- Cross-domain validation checklist
- Platform readiness and exit criteria
- Common terminology and interface assumptions
- Reference deployment and serviceability practices